

Ans. 1

Operating System is Software that help users and hardware communicate. It control CPU, memory and I/O devices. main goal are making system easy and making performance better. OS gives interface, file system and process handling. It also provide security and control resources. Use it is important because without OS application cannot run properly.

Ans. 2

Process and Thread are both execution unit. process is complete programme with own memory. Thread is smaller part of process and share memory with other thread. Thread operation are faster than process operation. process switching is heavy because separate context, but thread are lighter and better for multitasking inside same application.

Ans. 3

CPU Scheduling choose which process get CPU. FCFS work in arrived order and is easy to implement but if first process is large, other wait too much. Round robin give fixed time slice to each process and rotate. It is more fair and good for interactive system. However, it has more context switching, which can reduce efficiency if quantum is very small.

Ans. 4

Deadlock is when two or more process wait forever for resources. Coffman condition are mutual exclusion, hold and wait, no preemption and circular wait. If these all happen deadlock can occur.

Ex:

one process hold one lock and wait for second lock while another process hold second and wait for first.

Ans. 5

Virtual memory allow system to use disk as extra memory. Paging divides memory into pages and frames and map them using page table. It allow non-contiguous allocation and support larger programs. It reduce external fragmentation. But if too many page fault happens then system become slow.

Ans. 6

Synchronization is needed to avoid wrong result when many thread access same variable. Critical section should be protected. Mutex is lock / unlock mechanism for one thread at a time. Semaphore is counter based and can manage multiple resources also. These method prevent race condition and improve correctness.

Ans. 7

Fragmentation is memory wastage. Internal fragmentation is waste inside allocated block. External fragmentation is free memory splits in many small blocks. Internal can be reduced by better block sizing. External can be reduced by compaction and paging. paging is commonly used in modern operating system.